

Impact of Kernel Size & Stride on CNN Performance

Deep Learning

Group-12

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PROJECT OVERVIEW

PROBLEM STATEMENT

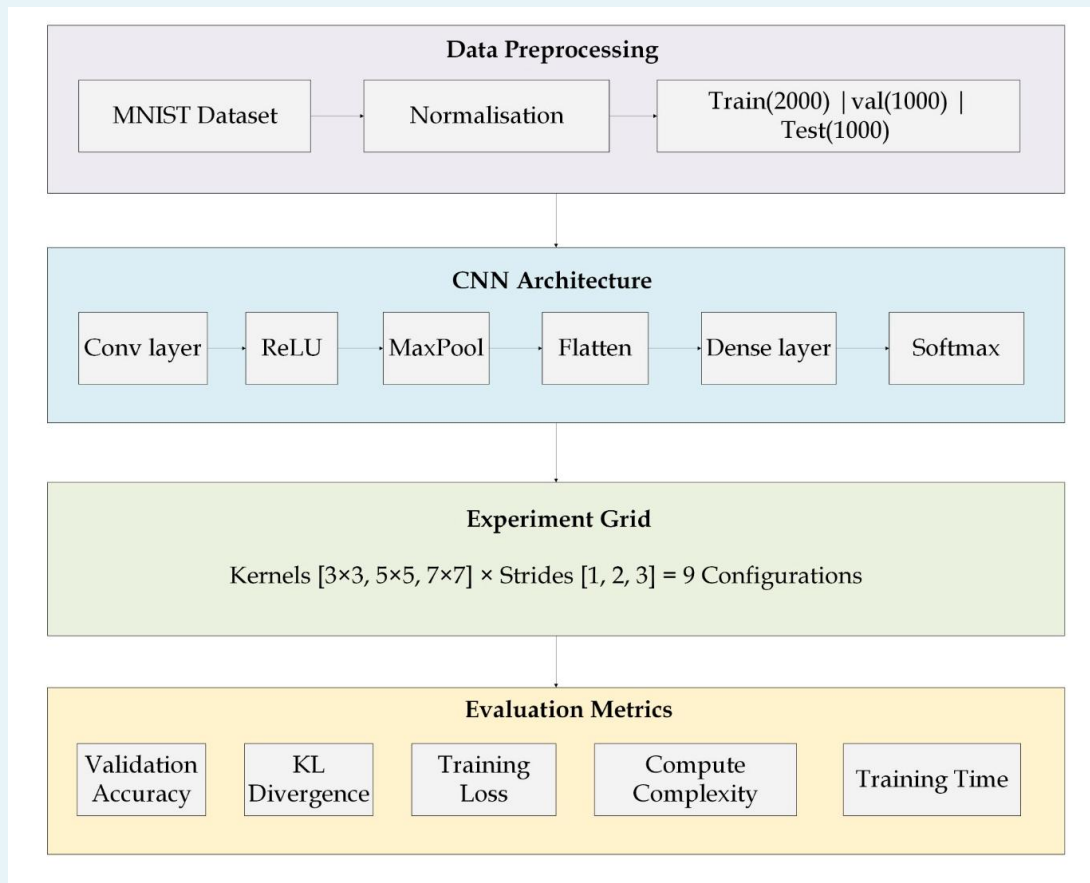
CNNs are widely used in image recognition tasks, but their performance is heavily influenced by architectural hyperparameters. Two of the most critical — kernel size and stride — are often selected by intuition or convention rather than systematic analysis.

For constrained settings like MNIST (28×28), how do these parameters interact to affect accuracy, prediction confidence (KL divergence), and computational cost? The trade-offs are not well understood at this scale.

OBJECTIVE

- Systematically evaluate all combinations of kernel sizes (3×3, 5×5, 7×7) and strides (1, 2, 3) — 9 configurations total
- Measure impact on validation accuracy, KL divergence, computational complexity, and training time
- Identify the optimal kernel–stride pair and understand the accuracy vs. efficiency trade-off
- Build the entire CNN from scratch using only NumPy — no deep learning frameworks

METHODOLOGY



TRAINING CONFIGURATION

Dataset	MNIST (28×28 grayscale)	Learning Rate	0.01
Train Samples	2,000	Optimizer	Mini-batch SGD
Validation Samples	1,000	Loss Function	Cross-Entropy
Test Samples	1,000	Kernels Tested	3×3, 5×5, 7×7
Epochs	10	Strides Tested	1, 2, 3
Batch Size	32	Total Configs	9 (independent runs)

COMPLETE RESULTS TABLE

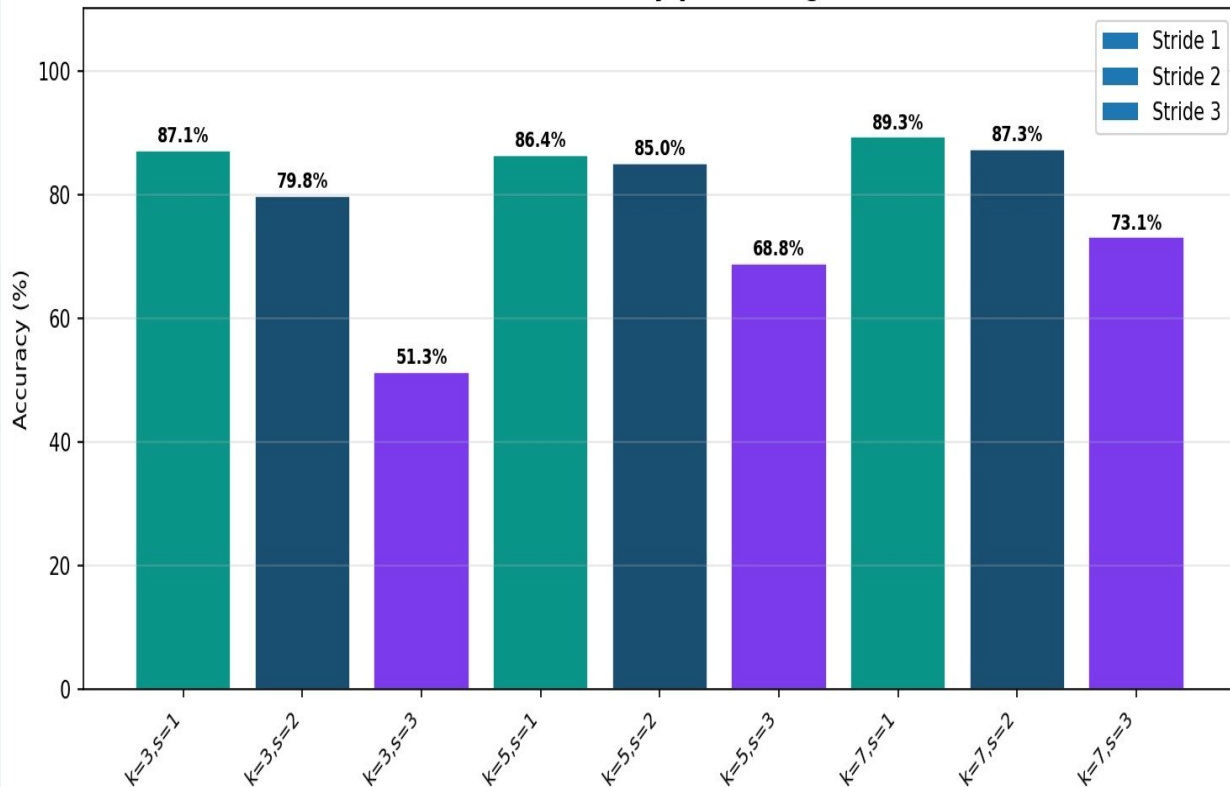
All 9 configurations | 10 epochs each

Kernel	Stride	Val Accuracy	KL Divergence	Complexity	Time (s)
3×3	1	0.8710	0.4269	48,672	2309.8
3×3	2	0.7980	0.6807	12,168	593.5
3×3	3	0.5130	1.7458	5,832	266.8
5×5	1	0.8640	0.4377	115,200	1989.1
5×5	2	0.8500	0.4870	28,800	522.1
5×5	3	0.6880	1.0918	12,800	222.0
7×7	1	★ 0.8930	★ 0.3823	189,728	1689.1
7×7	2	0.8730	0.4582	47,432	420.0
7×7	3	0.7310	0.8799	25,088	219.1

★ Best performing configuration

VALIDATION ACCURACY ANALYSIS

Validation Accuracy per Configuration



Key Observations



Stride=1 dominates

Highest accuracy for every kernel. Full spatial resolution preserved.



Stride=3 collapses

$3 \times 3, S=3 \rightarrow 51.3\%$. Too much spatial info discarded on 28×28 input.



7x7 leads overall

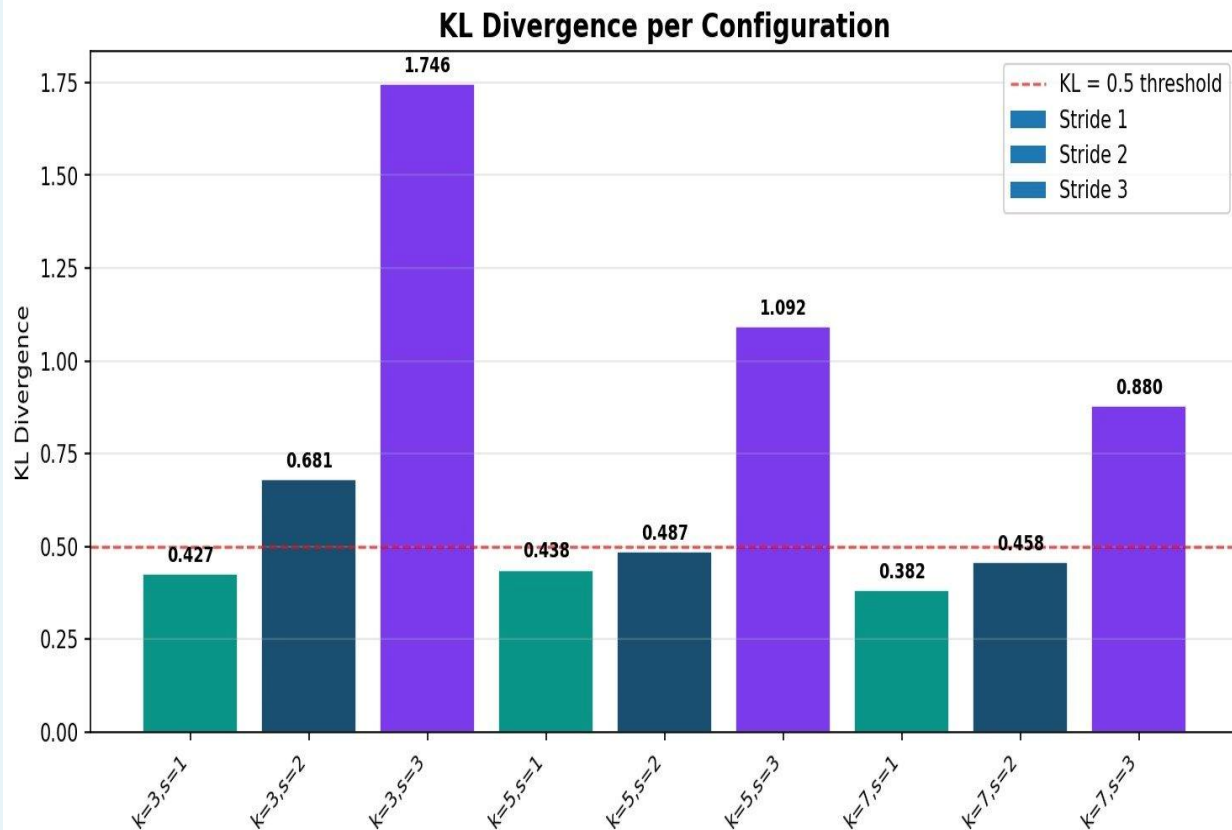
Best at every stride level. Broader receptive field captures more context.



5x5,S=2 sweet spot

85.0% accuracy at 4x lower complexity than $5 \times 5, S=1$.

KL DIVERGENCE ANALYSIS



Key Observations

7×7,S=1 best (0.382)

Sharpest, most confident predictions. Lowest distribution gap.

3×3,S=3 worst (1.746)

Near-random in probability space despite some correct argmax.

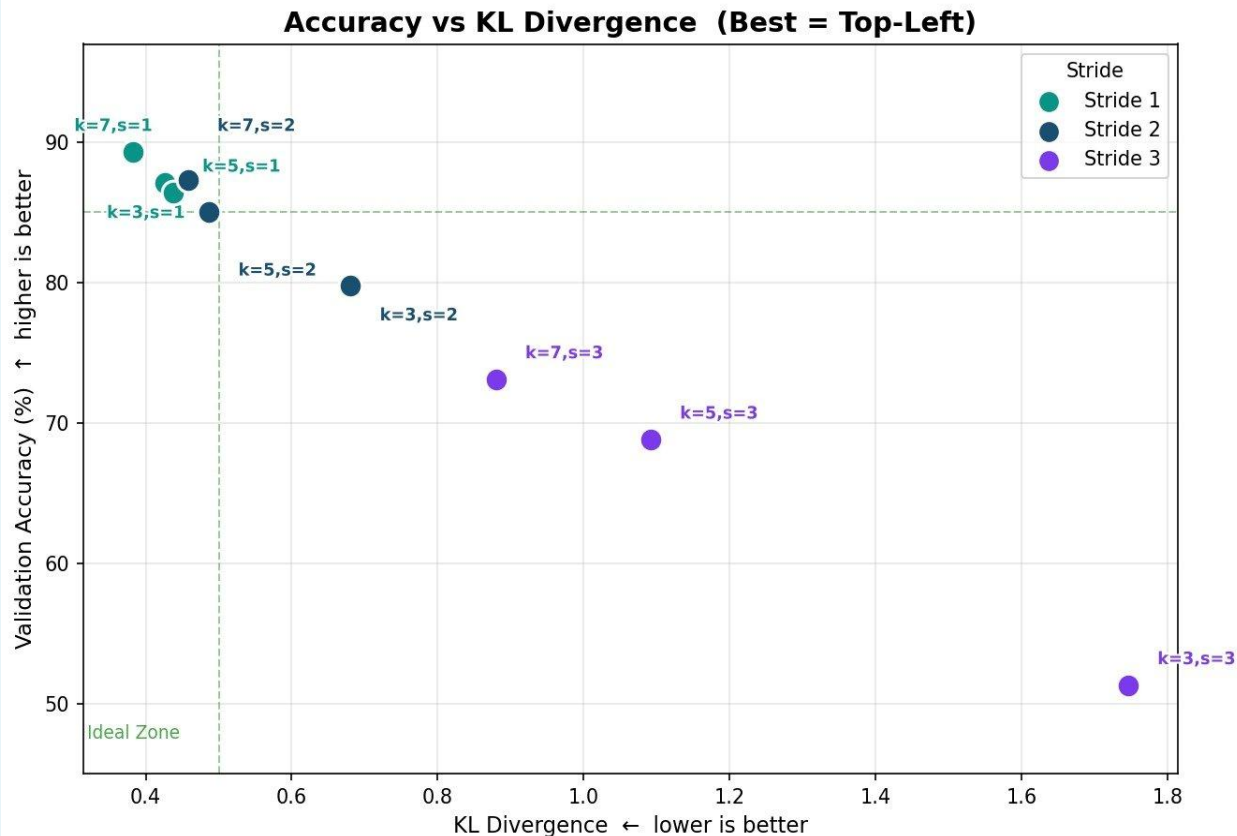
KL=0.5 threshold

Only stride=1 configs consistently stay below this confidence threshold.

Larger kernels help

KL decreases as kernel grows within every stride group.

ACCURACY vs. KL DIVERGENCE



Key Observations

Ideal zone: top-left

High accuracy + low KL = confident correct predictions. $7 \times 7, S=1$ leads.

Stride=3

All stride=3 configs cluster bottom-right — poor on both metrics.

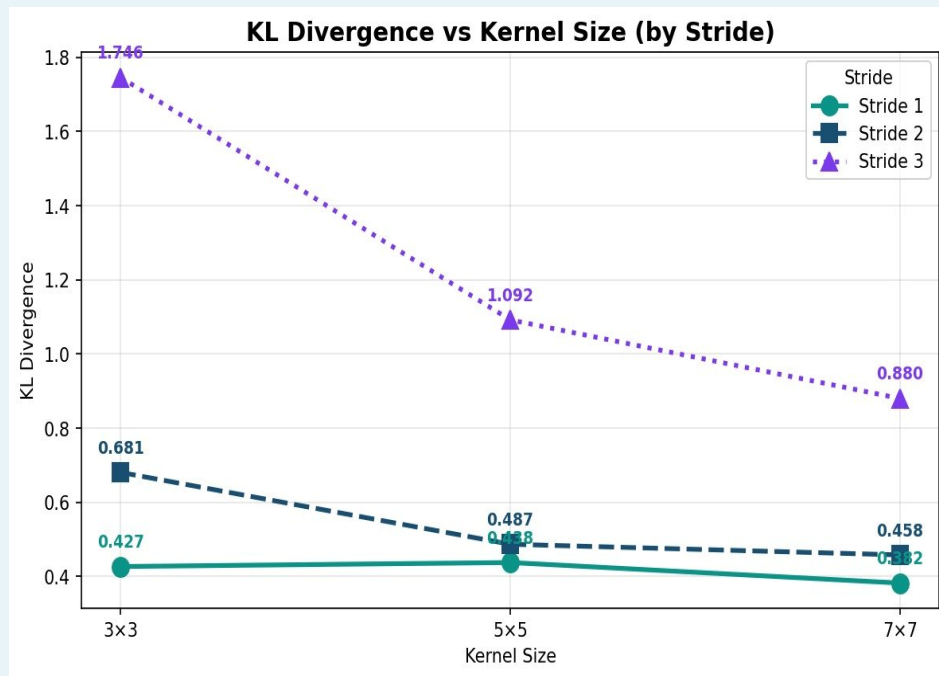
Stride=2

Stride=2 sits between stride=1 and stride=3 on both axes.

$k=5, S=2$

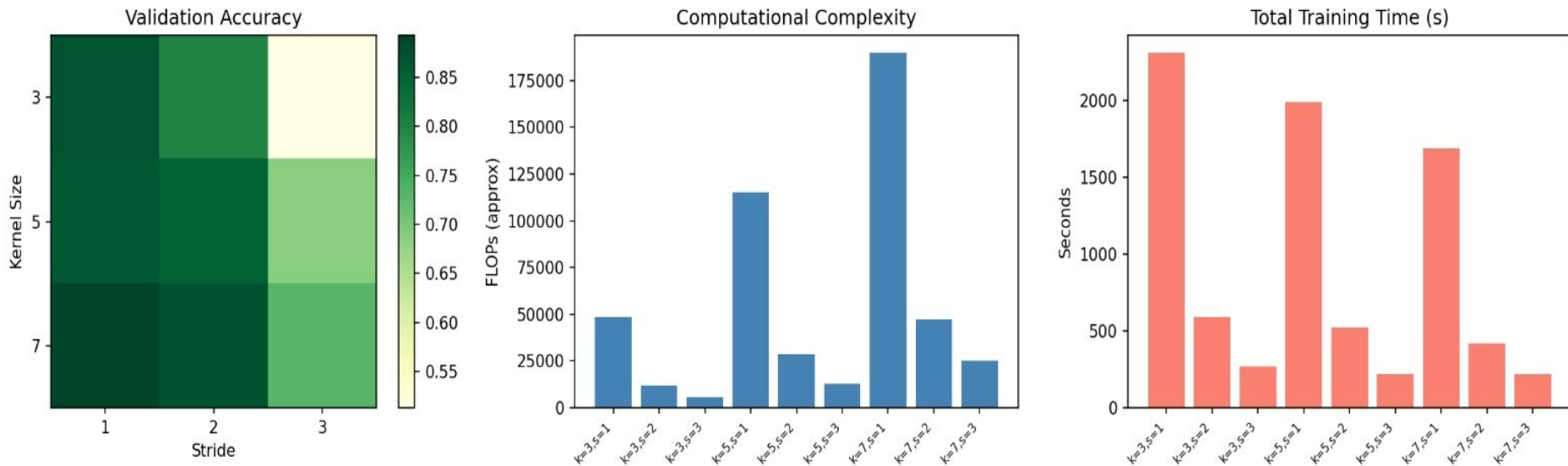
Beats $k=3, S=2$ on both accuracy and KL — best among stride=2.

COMPLEXITY, TIME & TREND ANALYSIS



- KL improves with larger kernels at every stride level.

COMPLEXITY, TIME & TREND ANALYSIS



- Heatmap confirms stride dominates accuracy
- complexity and time scale strongly with kernel size at stride=1

KEY FINDINGS

01

Stride is More Impactful Than Kernel Size

Increasing stride from 1 to 3 reduces accuracy across all kernel sizes, dropping as low as 51%. Keeping stride=1 consistently gives 87–89% regardless of kernel. Kernel size alone barely changes results within the same stride.

02

Larger Kernels Slightly Improve Accuracy

7×7 consistently outperforms 3×3 and 5×5 at every stride level. However, the gain over 5×5 is marginal (~0.3–1.5%) while complexity grows nearly 4x.

03

Complexity & Time Scaling Laws

Tripling stride reduces complexity 9× and training time ~7–10×. Doubling stride is the most practical trade-off: 75% complexity drop for only ~8% accuracy loss.

04

KL Divergence Tracks Accuracy Closely

Configurations with high accuracy also show sharper, more confident predictions (lower KL). Stride=3 models show $KL > 1.0$, near-random behavior in probability space despite some correct argmax predictions.

CONCLUSION

& Optimal Configuration

★ Recommended Configuration

Kernel: **7×7**

Stride: **1**

Val Accuracy: **89.3%**

KL Divergence: **0.3823**

⚡ Efficiency Pick (Time-Constrained)

7×7, Stride=2 — 87.3% accuracy at 420s training time (4× faster than 7×7, S=1). Best accuracy-to-compute ratio across all tested configurations.

Takeaways

- Stride is the dominant hyperparameter — prioritize stride=1 for accuracy-critical tasks.
- Kernel size provides moderate gains; 7×7 leads but requires ~4× more compute than 3×3.
- Stride=3 is too aggressive for 28×28 MNIST — spatial resolution collapses below the model's recovery threshold.
- For deployment trade-offs: 5×5, S=2 or 7×7, S=2 balance accuracy and speed well.
- KL divergence is a valuable companion metric — it reveals calibration quality beyond top-1 accuracy.